

ATTENDANCE SYSTEM USING RFID

Minor Project - II

Submitted

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Directorate of Polytechnic Education

CERTIFICATE

This is to certify that the report entitled “**ATTENDENCE SYSTEM USING RFID**” is submitted by “**K. Tharak Prabath (221FE05003)**” in the partial fulfilment of Minor Project – II work carried out in the Directorate of Polytechnic Education, VFSTR Deemed to be University.

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DECLARATION

I hereby declare that the project entitled " **ATTENDANCE SYSTEM USING RFID**" submitted for the "Directorate of Polytechnic Education". This dissertation is our original work, and the project has not formed the basis for the award of any degree, associateship and fellowship or any other similar titles and no part of it has been published or sent for publication at the time of submission.

By:

K. Tharak Prabath - 221FE05003

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With Sincere regards

K. Tharak Prabath - **(221FE05003)**

ABSTRACT

RFID technology is an automatic wireless identification system that works by the help of two components: a card and a reader. Using the RFID technology, the traditional system of taking attendance can be completely transformed to be digitalization. This system will help the authorities manage the attendance system in a more methodical, efficient and time saving manner. The proposed system has been implemented through a prototype that has proved the effectiveness of the concept in easing of taking attendance as a result of the automation due to the use of the RFID technology.

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INTRODUCTION

A Radio Frequency Identification (RFID) based attendance system is a system that uses tags to track when employees, students or any workers enter or leave an area. Radio Frequency Identification (RFID) technology is used to mark the attendance before every lecture in the institute. The data will be stored in the database. Here we use Excel sheet to store the data. Radio-frequency identification (RFID) is a technology that uses radio waves to transfer data from an electronic tag, called RFID tag or label, connected to an object, via a reader for the cause of identifying and monitoring the object. Radio frequency identification (RFID) is a matured technological knowhow that accommodates the use of electromagnetic or electrostatic coupling in the radio frequency portion of the electromagnetic spectrum to uniquely become aware of an object, animal, or person. RFID chips include a radio transmitter that emits a coded identification number when queried through a reader device. Some RFID tags can be examined from various meters away and beyond the line of sight of the reader. The utility of bulk reading allows an almost-parallel analysing of tags. This small type is incorporated in client products, and even implanted in pets, for identification.

The RFID tag includes a small RF transmitter which transmits an encoded radio sign to interrogate the tag, and receiver which receives the message and responds with its identification information. Some RFID tags do not use a battery. Instead, the tag makes use of the radio power transmitted by way of the reader as its strength source. The RFID machine plan consists of an approach of discriminating countless tags that might be within the range of the RFID reader.

RFID can be used in many applications. A tag can be affixed to any object and used to track and manage inventory, assets, people, etc. For example, it can be affixed to cars, laptop equipment, books, cellular phones, etc.

Components Description

We had been used five components in this project. Let we have a brief detailed description of each of the components. The Components used for this project have been mentioned below:

1. Arduino UNO
2. RFID module
3. RFID tag or card
4. Bread board
5. Jumper Wires
6. LCD display
7. Resistors
8. LED light

- **Arduino UNO:**

In the name Arduino UNO, UNO suggests that ‘one’ in Italian. It was labelled as UNO as it was the first released micro-controller board by the Arduino software. It is a micro-controller board based on the ATmega328P.

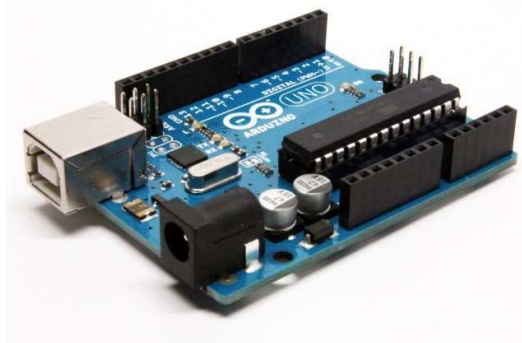


Fig 1: Arduino UNO

The board Arduino UNO is having the total of 28 pins. Out of the 28 pins, 14 pins are the digital input/output pins (of which 6 can be used as the PWM outputs). There are also 6 analog pins. The input voltage varies from the 7V to 20V.

The Arduino UNO has a number of facilities for communicating with a computer, another Arduino board, or other micro-controller. The ATmega328 provides UART TTL (5V) serial Communication, which is available on digital pins 0 (RX) and 1(TX). An ATmega16U2 on the board channels this serial communication over USB and appears as a virtual com port to software on the computer. The 16U2 firmware uses the standard USB COM drivers, and no external driver is needed. However, on windows, a .inf file is required. Arduino software (IDE) includes a serial monitor which allows the simple textual data to sent to and from the board. The RX and TX LEDs on the board will flash when data is being transmitted via USB-to-serial chip and USB connection to the computer (but not serial communication on pins 0 and 1). A software serial library allows the serial communication on any of the UNO's digital pins.

- RFID Module

The term RFID is meant by the Radio Frequency Identification, as the name defines the operation of the device is based on the Radio frequency signals. The RFID systems consists of RFID reader and a tag or card which is normally, used in the identification and tracking of objects. Similar to barcode the RFID can also give unique identification number to all products but the added advantage is unlike the barcode system's line of sight, this system can detect the RFID tag within its proximity range.

you do not need a human to search for the barcode and point the barcode scanner on it.

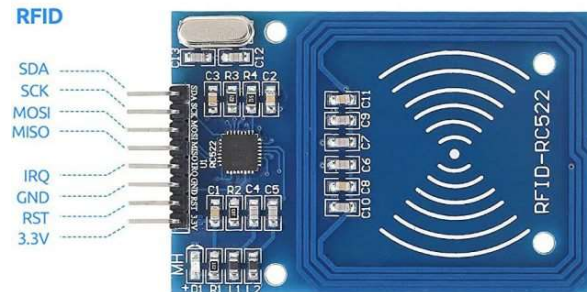


Fig 2: Radio Frequency Identification (RFID)

Every RFID consists of the three components: a scanning antenna, a transceiver and a transponder. When the scanning antenna and transceiver are combined, they are referred to as an RFID reader or interrogator. There are two types of RFID readers – fixed reader and mobile readers. The RFID reader is a network connected device that can be portable or permanently attached. It uses the Radio waves to transmit the signals that activate the tag. Once activated, the tag sends the wave back to the antenna, where it is translated into data.

- RFID Tag

RFID tags are small electronic devices that uses the radio waves to transmit data. They consist of an antenna and a microchip that stores the information. These tags are used to identify and track objects, people, or animals.

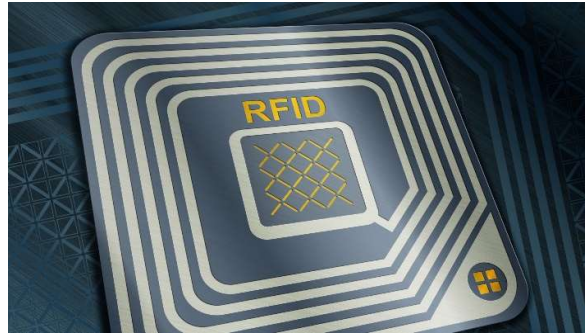


Fig 3: RFID Tag

The Radio Frequency Identification (RFID) tag contains a unique identifier, such as a product number or serial number.

An RFID reader emits the radio waves that activate the tag's antenna. The tag's antenna receives the radio waves and transmits the stored data to the sender. The reader decodes the data and sends it to a computer system to further processing.

There are two types of RFID tags. They are

1. Passive RFID tags:

These tags do not have their own power source. They receive power from the reader's radio waves to activate and transmit the data. They are commonly used in retail and supply chain management.

2. Active RFID tags:

These tags have their own power source, usually a battery. They can transmit data over longer distances and at higher frequencies than passive tags. They are used in the applications that require the frequent updates or long-range tracking, such as asset tracking and live stock management.

- Bread board

A breadboard (sometimes called a plug-block) is used for building temporary circuits. It is useful to designers because it allows components to be removed and replaced easily. It is useful to the person who wants to build a circuit to demonstrate its action, then to reuse the components in another circuit.

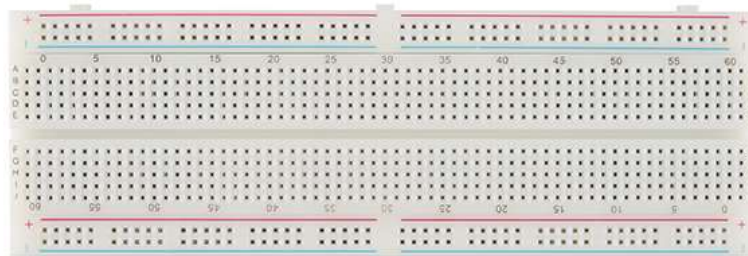


Fig 4: Bread Board

- Jumper wires

Jumper wire is a conducting wire that is used to transfer the electrical signals between two points in a circuit. In breadboards, jumper wires are used to establish connections between the central controller and other devices such as buttons and sensors. Jumpers can be say as connectors. It allows the to connect the two points with each other without using soldering.



Fig 5: Jumper Wires

- LCD Display

A liquid crystal display or LCD draws its definition from its name itself. It is a combination of two states of matter, the solid and the liquid. LCD uses a liquid crystal to produce a visible image. Liquid crystal displays are super-thin technology display screens that are generally used in laptop computer screens, TVs, cell phones, and portable video games. LCD's technologies allow displays to be much thinner when compared to a Cathode Ray Tube (CRT) technology.



Fig 6: Blue 16x2 LCD display

- Resistors

A Resistor is a passive component that opposes the flow the electrons. The main purpose of the resistor is to reduce the current flow or lower the voltage in any circuit. It is made of copper wires which are coiled around a ceramic rod and outer part of the resistor is insulated with an isolating paint.



Fig 7: Resistor

- LED light

A light-emitting diode (LED) is a semiconductor device that emits light when an electric current flows through it. When current passes through an LED, the electrons recombine with holes emitting light in the process. LEDs allow the current to flow in the forward direction and blocks the current in the reverse direction.



Fig 8: Light Emitting Diode

Block Diagram

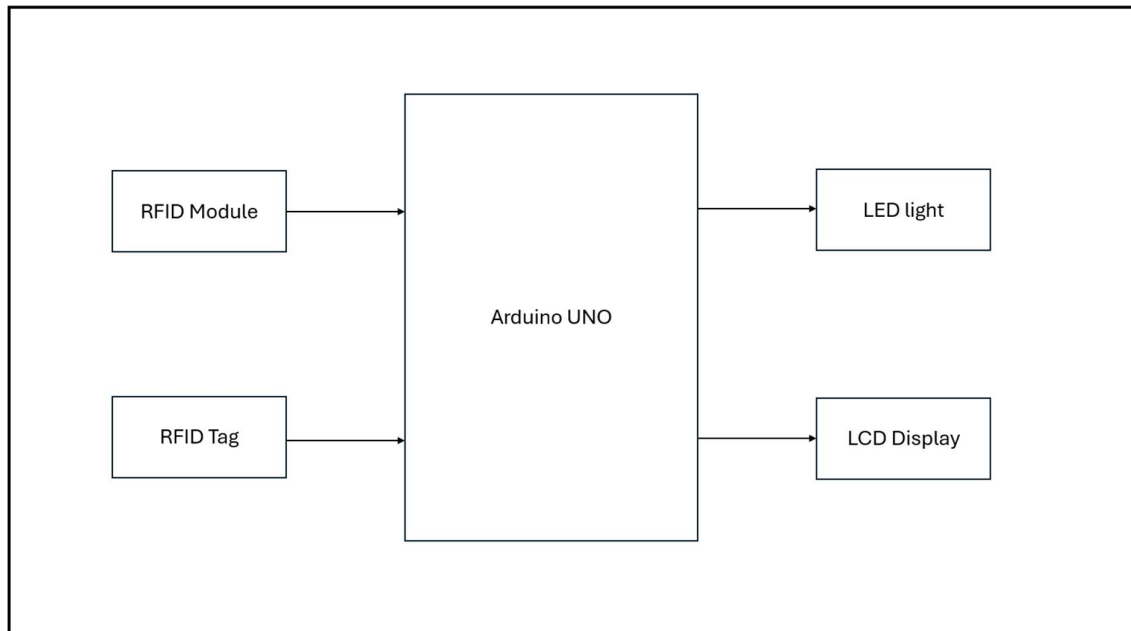


Fig 9: Block Diagram

Overall Function:

This system appears to be designed for access control or authentication purposes. It likely uses an RFID tag to identify a person or object, and based on the tag's data, it triggers an LED light and displays information on an LCD screen.

Power Supply: The Arduino, RFID module, LED, and LCD display will need a suitable power supply, likely a 5V DC power source.

Communication: The Arduino will need to communicate with the RFID module and the other components. This is often done using serial communication or specific protocols for RFID modules.

Programming: The Arduino will need to be programmed with code to handle the RFID data, control the LED, and display information on the LCD. This programming is typically done using the Arduino IDE

Circuit and Program

The circuit diagram of the RFID attendance system is mentioned below:

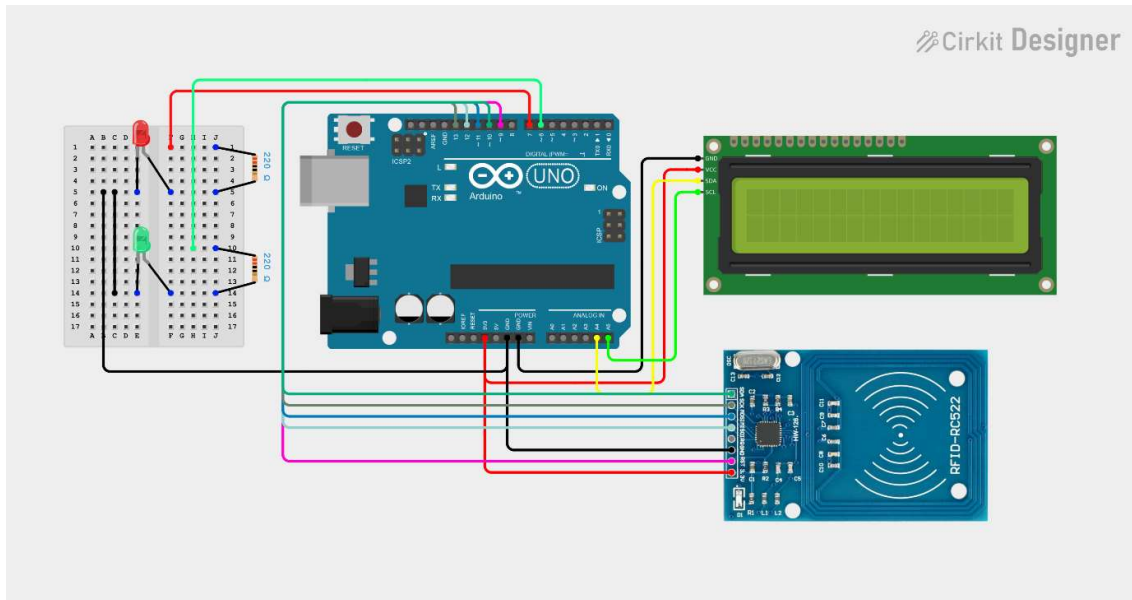


Fig 10: Circuit diagram of RFID system

Program

```
#include <SPI.h>
#include <MFRC522.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#define SS_PIN 10
#define RST_PIN 9
#define LED_DENIED_PIN 7
#define LED_ACCESS_PIN 6
LiquidCrystal_I2C lcd(0x3F,16,2);
MFRC522 mfrc522(SS_PIN, RST_PIN); // Instance of the class
```

```

    int code[] = {32,154,149,117}; //This is the stored UID (Unlock
Card)

    int codeRead = 0;

    String uidString;

    void setup() {
        Serial.begin(9600);
        SPI.begin();    // Init SPI bus
        mfrc522.PCD_Init(); // Init MFRC522
        Serial.println("Arduino RFID reading UID");
        pinMode( LED_DENIED_PIN , OUTPUT);
        pinMode( LED_ACCESS_PIN , OUTPUT);
        lcd.init();

        // Print a message to the LCD.
        lcd.backlight();
        lcd.setCursor(0,0);
        lcd.print("Show your card:");
    }

    void loop() {
        if ( mfrc522.PICC_IsNewCardPresent())
        {
            if ( mfrc522.PICC_ReadCardSerial())
            {
                lcd.clear();
                Serial.print("Tag UID:");
                lcd.setCursor(0,0);
            }
        }
    }

```

```

        lcd.print("Tag UID:");
        lcd.setCursor(0,1);
        for (byte i = 0; i < mfrc522.uid.size; i++) {
            Serial.print(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : "
");
            Serial.print(mfrc522.uid.uidByte[i], HEX);
            lcd.print(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " ");
            lcd.print(mfrc522.uid.uidByte[i], HEX);
            // lcd.print(" ");
        }
        Serial.println();
        int i = 0;
        boolean match = true;
        while(i<mfrc522.uid.size)
        {
            if(!(int(mfrc522.uid.uidByte[i]) == int(code[i])))
            {
                match = false;
            }
            i++;
        }
        delay(3000);
        lcd.clear();
        lcd.setCursor(0,0);
        if(match)

```

```

        {
            digitalWrite( LED_ACCESS_PIN , HIGH);
            lcd.print("Authorized access");
        }else{
            digitalWrite( LED_DENIED_PIN , HIGH);
            lcd.print(" Access denied ");
            Serial.println("\nUnknown Card");
        }

Serial.println("=====");
        mfrc522.PICC_HaltA();
        delay(3000);
        reset_state();
    }
}
}

void reset_state()
{
    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("Show your card:");
    digitalWrite( LED_ACCESS_PIN , LOW);
    digitalWrite( LED_DENIED_PIN , LOW);
}

```

Working

The working of the RFID based attendance system working is below:

1. **RFID Tag:** Each individual is assigned a unique RFID tag, which is typically a small, passive electronic device. This tag contains a microchip that stores a unique identification number.
2. **RFID Reader:** RFID readers are installed at designated locations, such as entrances and classrooms. These readers emit radio waves that activate the RFID tags within their range.
3. **Data Transmission:** When an individual with an RFID tag comes within the range of a reader, the tag's unique identification number is transmitted to the reader.
4. **Data Processing:** The reader then sends this information to a central computer system. The system compares the received identification number with a database of authorized individuals.
5. **Attendance Recording:** If the identification number matches a record in the database, the system records the individual's attendance, including the time and location of the scan.
6. **Data Storage and Analysis:** The recorded attendance data is stored in a database for further analysis and reporting. This data can be used to generate attendance reports, identify trends, and track individual attendance patterns.

There is a coil present in the tag or card and it generates the electromagnetic field which couple into the coil on the RFID tag. RFID card is detected by the reader module if it in the range of 10cm. After detects the card or tag it checks that is original or fake. If it is the original card then the information is sent to the data base. Then the information is stored to the data base. When all the cards are scanned and we can save the file in the data base and close the file. Here we use data base as the Excel work sheet.

Advantages

There are some of the key advantages of RFID based attendance system. The advantages are mentioned below:

1. Accuracy:

RFID technology eliminates the potential for human error and ensures accurate attendance records.

2. Efficiency:

The automated process significantly reduces the time and effort required for manual attendance taking.

3. Security:

RFID tags can be programmed with additional security features, such as encryption, to protect sensitive information.

4. Real-time tracking:

The system provides real-time tracking of individuals' movements within a facility.

5. Integration with other systems:

RFID-based systems can be integrated with other software systems, such as HR and student information systems, for comprehensive data management.

Disadvantages

Here are some of the limitations of the RFID based attendance system.

1. High initial cost:

RFID systems require significant upfront investment in hardware (readers, tags) and software. This can be particularly expensive for large organizations.

2. Cost of tag replacement:

RFID tags have a limited lifespan and need to be replaced periodically, adding to the ongoing cost of the system.

3. Technical complexities:

RFID systems can be complex to set up and maintain, requiring specialized technical expertise.

4. Potential for interface:

RFID signals can be susceptible to interference from other electronic devices or environmental factors, which may affect the system's accuracy.

5. Security Concerns:

While RFID systems can be secure, there are potential risks of unauthorized access or data breaches if the system is not properly implemented and protected.

6. Lack of Privacy:

Some individuals may have concerns about the privacy implications of RFID technology, as it can track a person's movements and activities.

Applications

There are some common applications are mentioned in this project.

1. Schools and Universities:

Tracking student attendance in classrooms and laboratories.

2. Workplaces:

Monitoring employee attendance and shift timings.

3. Access Control:

Controlling access to restricted areas within a facility.

4. Event Management:

Tracking attendance at conferences, seminars, and other events.

Conclusion

This project is a full system for student monitoring in colleges, and higher educational institutions. Lectures no need to waste their time to take their attendance. They do not need to carry the attendance registers. Attendance will directly store in the data base.

The RFID-based attendance system represents a significant advancement over traditional manual methods of attendance tracking. By leveraging Radio Frequency Identification (RFID) technology, this system offers a highly efficient, accurate, and automated way to monitor the students attendance in the educational institutions.

The implementation of an RFID-based attendance system presents a significant advancement in the management of attendance records. By automating the attendance process, it minimizes human errors and administrative burdens, ensuring reliable and timely data collection.

The RFID-based attendance system is a robust and scalable solution that aligns with the evolving needs of modern organizations.

Future Scope

Information will directly added to the data base:

The data will store directly to the excel spread sheet when the tag is connected.

Enhanced security:

The security for the information or records of attendance will store safely.

Removing of tag:

The electromagnetic coil which has present in the tag will remove and it will add to the college ID card. So there will no need carry the tag.

Biometric scan may available:

To reduce the unauthorized scan we can implement the biometric. By this, we can able to make an fingerprint instead of carrying the college ID.